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Research Article submission

Dear Editors of IEEE Access,

Please consider our manuscript, “Protocol-Constrained Residual Adaptation with Frozen Vision–Language Features for Rigid-Body Inference,” by Hyojin Park, Nam-Hyun Yoo, and Jinhong Yang, for publication as a **Research Article**.

The study examines whether a compact visual residual can add predictive information after a strong numerical estimator has already used a short motion observation. The method combines fixed robust estimation, numerical residual adaptation, and a bounded frozen-feature residual with protocol-specific permissions. Its contribution is the tested allocation of roles and the measured increment, rather than a claim that hybrid estimation or VLM-based physical inference is new in isolation.

On 72 new synthetic geometry families, the complete method reduces canonical analytic new-action error by 29.95% relative to the strongest fixed robust point baseline. Most of the improvement comes from numerical adaptation; the visual increment is 1.07%, with one-sided support and a two-sided interval containing zero. A separate offline public simulator task yields a 4.27% visual-plus-gate increment after task-specific development fitting. The manuscript distinguishes this result from zero-shot transfer and reports conservative empirical coverage and the unresolved real-video transfer result.

We believe that the combination of structured estimation, controlled visual-feature use, and inspectable evaluation is relevant to the applied artificial intelligence and engineering readership of IEEE Access. The accompanying materials include manuscript source, figures, supplementary implementation notes, and a portable replay of saved paired statistics. The public-task selection and group-independence limitations are stated explicitly. AI assistance and research funding are disclosed in the manuscript.

The supplement also reports cached-feature component experiments, a mean-fixed covariance sensitivity analysis, and initialization-Jacobian diagnostics. These analyses retain the original model and are explicitly distinguished from independent confirmation.

A further new-episode study separates both object and surface friction conditions during refitting. On 329 evaluation episodes, it retains a 4.38% material-pair improvement with a crossed-material 95% interval of $[-6.044, -0.116]$ mm. The manuscript distinguishes this evidence from new object/scene-asset transfer and discloses historical architecture selection within the material bank. Versioned verification materials are provided through the linked public GitHub repository.

Thank you for considering this work.

Sincerely,
Jinhong Yang
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